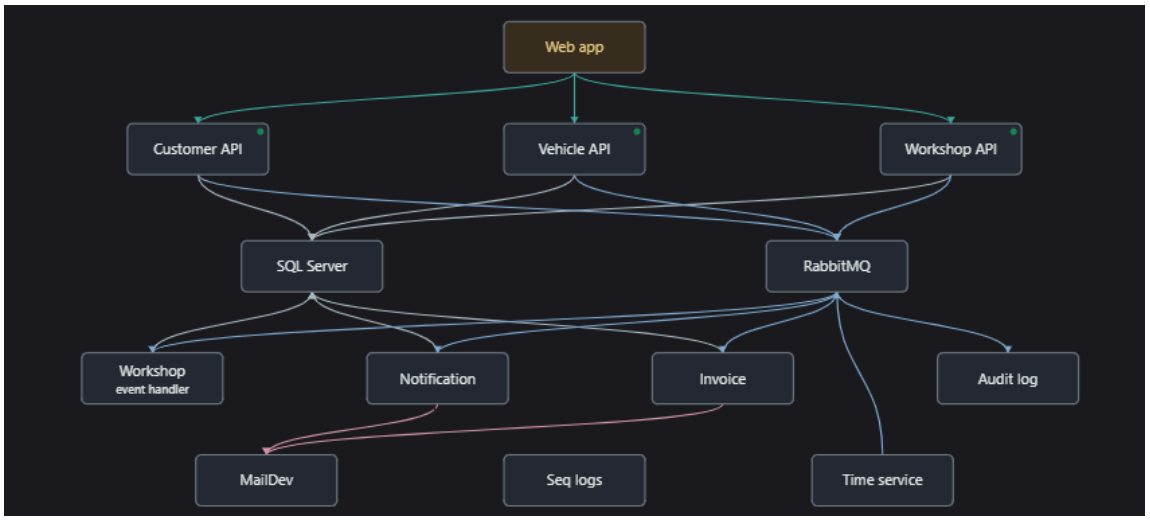
Pitstop Chaos Agent

Team No. 28 | AI-driven chaos engineering in an isolated Docker sandbox

Application Dependencies



Web app → three HTTP APIs → SQL Server; domain events travel through RabbitMQ to Workshop projection, invoice, notification and audit workers. Time service emits daily events; mail and Seq support delivery and logs. Map is semi-automatically documented and manually reviewed.

Decision And Evidence Pipeline

Baseline + incident history + dependency map → Discovery worker

Coordinator → AI planner → allowlisted Injection worker + rollback

Detection worker → AI Recommendation worker → evidence-linked findings

Independent live monitor → dashboard, JSON / CSV and ranked report

Component	Responsibility
Specialized workers	Isolated discovery, planner, injection, detection and recommendation processes. Planner and recommendations share Qwen; deterministic workers measure and execute. Unimplemented failure ideas remain proposals.
Safety coordinator	Local Docker and checkout ownership; healthy preflight; one fault at a time; duration watchdog; unrelated-module / blast-radius thresholds; STOP; verified recovery gate. No arbitrary model commands.
Experiment tools	API, SQL or broker crash; 500/1000 ms Workshop delay. Combined: API crash with two read-load connections, 10 GETs/sec, 30 sec maximum. Stronger delay: 20 sec maximum.
Dashboard / monitor	Loopback-only server; live stages, worker PIDs, metrics, 2D/3D dependencies and fixes. Independent read-only detection and recovery tracking. Confirmed runs and emergency STOP.

Scope And Limits

Container restart restores services without deleting database volumes. Read probes detect HTTP-200 offline fallbacks but do not prove write correctness or message durability. AI findings are hypotheses for human review. The score is a disclosed experiment-specific heuristic, not an AI-certified SLA. No production or shared systems are targeted.

Source: saurabhshreni-cmyk/bosch-chaos-hackathon | Application license: Apache-2.0 | Authoritative connections: agent/dependencies.json